particles with definite positions into the theory, which accounted for the outcomes of measurements. Everett didn't add particles—he didn't think he needed them. Instead, he insisted that a single universal wave function was all there was: a massive mathematical object describing the quantum states of all objects in the entire universe. This universal wave function, according to Everett, obeyed the Schrödinger equation at all times, never collapsing, but splitting instead. Each experiment, each quantum event, spun off new branches of the universal wave function, creating a multitude of universes in which that one event had every possible outcome. Everett's shocking idea came to be known as the "many-worlds" interpretation of quantum physics.

The many-worlds interpretation sounds absurd at first blush, and probably at second blush too. We live in one world, not a multitude of them. If every quantum event—and in a fully quantum world, that's every event of any kind—leads the universe to split, where are these other universes? How could there possibly be so many of them without any indication that they're there? And, for that matter, how can any single event—one photon going through one double-slit apparatus, for example—cause the entire universe to split? To understand how the many-worlds interpretation accounts for these problems, let's take a second look at a simple quantum experiment, even simpler than the double slit: Schrödinger's cat.

Way back in the Introduction, we met Schrödinger's thought experiment, which has given the ASPCA nightmares for over eighty years. Put a cat in a box, along with a vial of poison and a lump of weakly radioactive metal; set up a Geiger counter (radiation detector) and a hammer so that the hammer will smash the vial if the detector detects any radiation. Then leave the cat in the box long enough that there's a 50-50 chance that the metal has emitted radiation. Now what? Is the cat dead or alive? According to the Copenhagen interpretation, the question is meaningless—you can't ask what happened before you open the box, because it's unobservable. According to Bohm and the pilot-wave interpretation, the question is quite meaningful, we just don't know the answer to it. The cat is either dead or